Camille Yabu

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Education

University of California, Santa Cruz — Santa Cruz, CA

M.S. Computer Science and Engineering, Contiguous BS/MS Pathway (applying) — *Expected June 2028*

B.S. Computer Engineering, Systems Programming Concentration, Computer Science Minor — *Sept. 2023 – June 2027*

- Relevant Coursework: Data Structures & Algorithms, Computer Systems Design, Computer Architecture, Software Engineering, Embedded System Design, Machine Learning Basics, Linear Algebra, Signals and Systems, Database Systems (current), Parallel & Concurrent Programming (current)

Experience

AI Software Engineer Intern — June 2026 – August 2026

Sierra Space — *Louisville, CO*

- Architected and deployed an enterprise AI agent translating natural language to SQL across production databases, optimizing from a high token prototype to a production system with **under 2 second response time**
- Engineered a semantic schema filtering algorithm and NoSQL caching pipeline that intelligently selects relevant database tables per query, **reducing LLM token consumption by 97%** while maintaining **100% accuracy**
- Researched authentication protocols and implemented an **On Behalf Of token exchange flow** with identity preserving middleware, enabling per user security enforcement and audit logging for cloud deployment
- Built **3 production Model Context Protocol servers** with **OAuth 2.0** and async operations, standardizing how AI agents access collaboration tools, productivity suites, and databases
- Trained a **PyTorch EfficientNet B2** model on PCB images with focal loss for class imbalance, benchmarked against Azure Custom Vision, and delivered a feasibility report with statistical analysis on dataset needs

Software Developer — October 2025 – June 2026

ReCat

- Built and shipped **React Native** and **TypeScript** features from Figma designs and Linear tickets
- Integrated **Firebase Authentication** and **Firestore** for cloud backed user sessions and persistent data

AI Research Fellow — January 2026 – April 2026

Handshake

- Evaluated AI outputs across audio, visual, and text tasks, setting a quality baseline that guided model improvements
- Developed golden solutions and adversarial test cases for complex software tasks, uncovering evaluation gaps in LLM performance

Projects

CircleHome: Household Coordination Mobile App | *Dart, Flutter, Firebase, Scrum* — June 2026

- **Product Owner** on a five person Scrum team, leading sprint planning, backlog prioritization, and demos
- Architected task creation with difficulty scaling, recurring options, and Firestore completion tracking with undo
- Built a toggle between My Tasks and All Tasks views with filtered Firestore queries and real time feed updates

Phish-Or-Friend: Cybersecurity Education Game | *Luau, Rojo, Python, Fetch.ai* — April 2026

- Built a Roblox phishing game using **Fetch.ai multi agent architecture** to generate scam and safe scenarios
- Implemented a Python backend with Fetch.ai uAgents enabling adaptive difficulty across **5+ scenario categories**
- Split the codebase into a **Luau/Roblox client** and a real time **Python backend**

ML Based Stock Return Prediction System | *Python, scikit-learn, TensorFlow, Alpaca API* — March 2026

- Built an end to end ML pipeline predicting next day returns for AAPL and NVDA using **19 engineered features**
- Benchmarked **Random Forest, LSTM, and Transformer encoder** models across a unified feature pipeline

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C, C++, Java, Dart, Verilog, Bash, HTML/CSS

AI/ML & Cloud: Claude Agent SDK, OpenAI API (GPT-4o), AWS Bedrock, PyTorch, TensorFlow, scikit-learn, MCP, RAG, LLM agents, LLM evaluation, LangChain, LangGraph, prompt engineering, Azure (OpenAI, SQL, Cosmos DB, Custom Vision, AD)

Frameworks & Tools: React Native, Flutter, Firebase, Flask, FastAPI, Node.js, Fetch.ai, Docker, CI/CD, Git, GitHub, Jupyter, Vivado

Systems, Auth & Data: PostgreSQL, MySQL, Linux, multithreading and concurrency, OAuth 2.0, OBO auth flow, REST API design, NumPy, pandas, OpenCV, Matplotlib, FPGA (BASYS3)